

Unit - 1 : Object Oriented Thinking and Java Basics.

Introduction

→ Object Oriented through programming (OOP) is a programming method that Organizes programs into Objects

→ An Object contains data and methods (functions)

→ Objects Communicate by sending messenger.

Features of OOPS.

1) Classes

→ Collection of Objects of classes.

2) Objects

→ A real-world Entity that contains data & methods.

→ It also, Guides, Supports a Community.

Ex: Class Test

keyword class name

class: Flower

Rose

Jasmine

Sunflower

} This are objects of flowers.

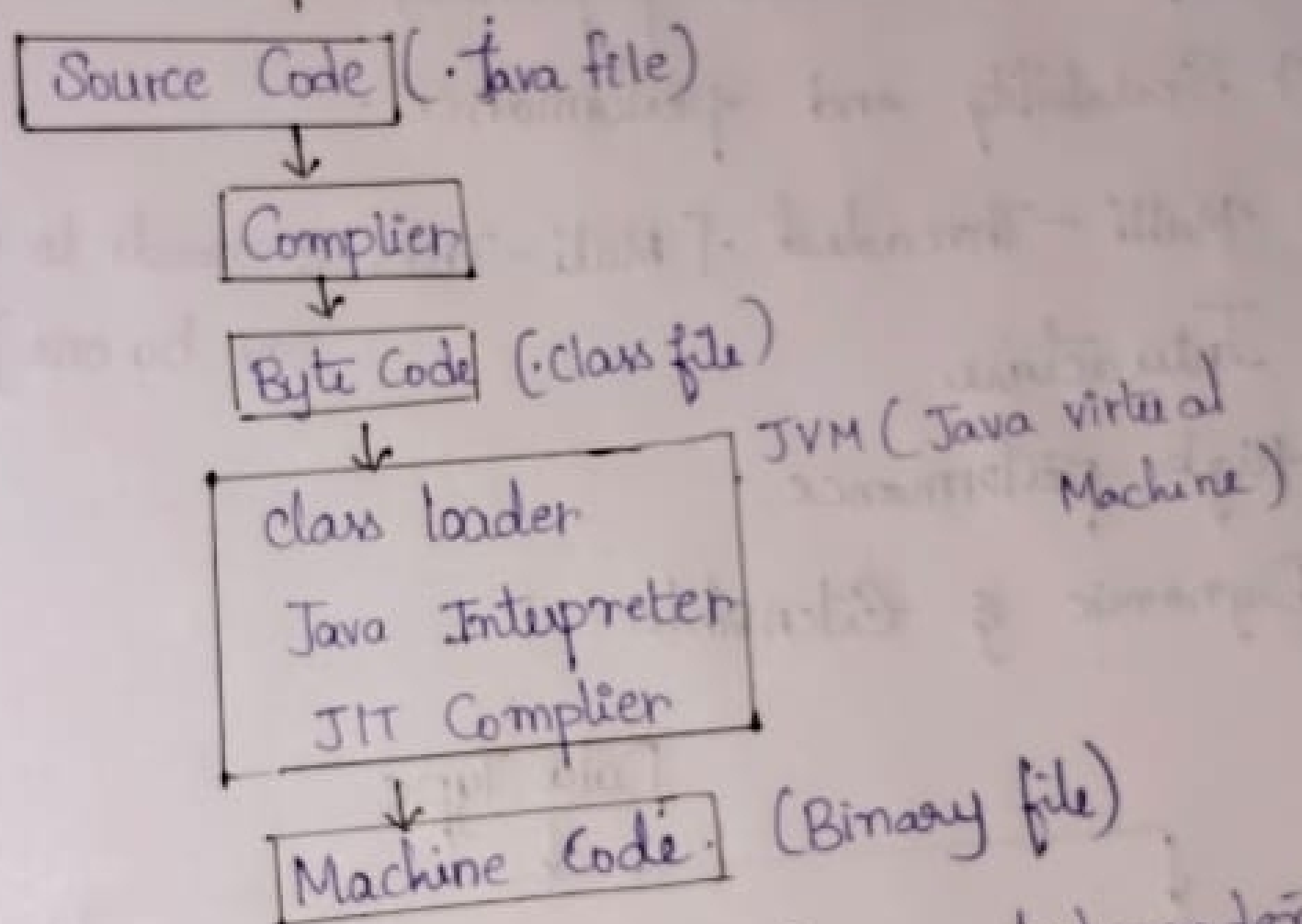
→ C programming is topdown approach.

→ whenever you write class name the first letter should be in Capital.

History of Java

- Developed by James Gosling and team at Sun microsystem in 1991.
- At that time it is used call as 'Oak' not as 'Java'
- It is used for Consumer products like microwave.
- Developing on Internet application was from 1993-1995 and named as Java in 1995.

★ Why Java is platform Independent?



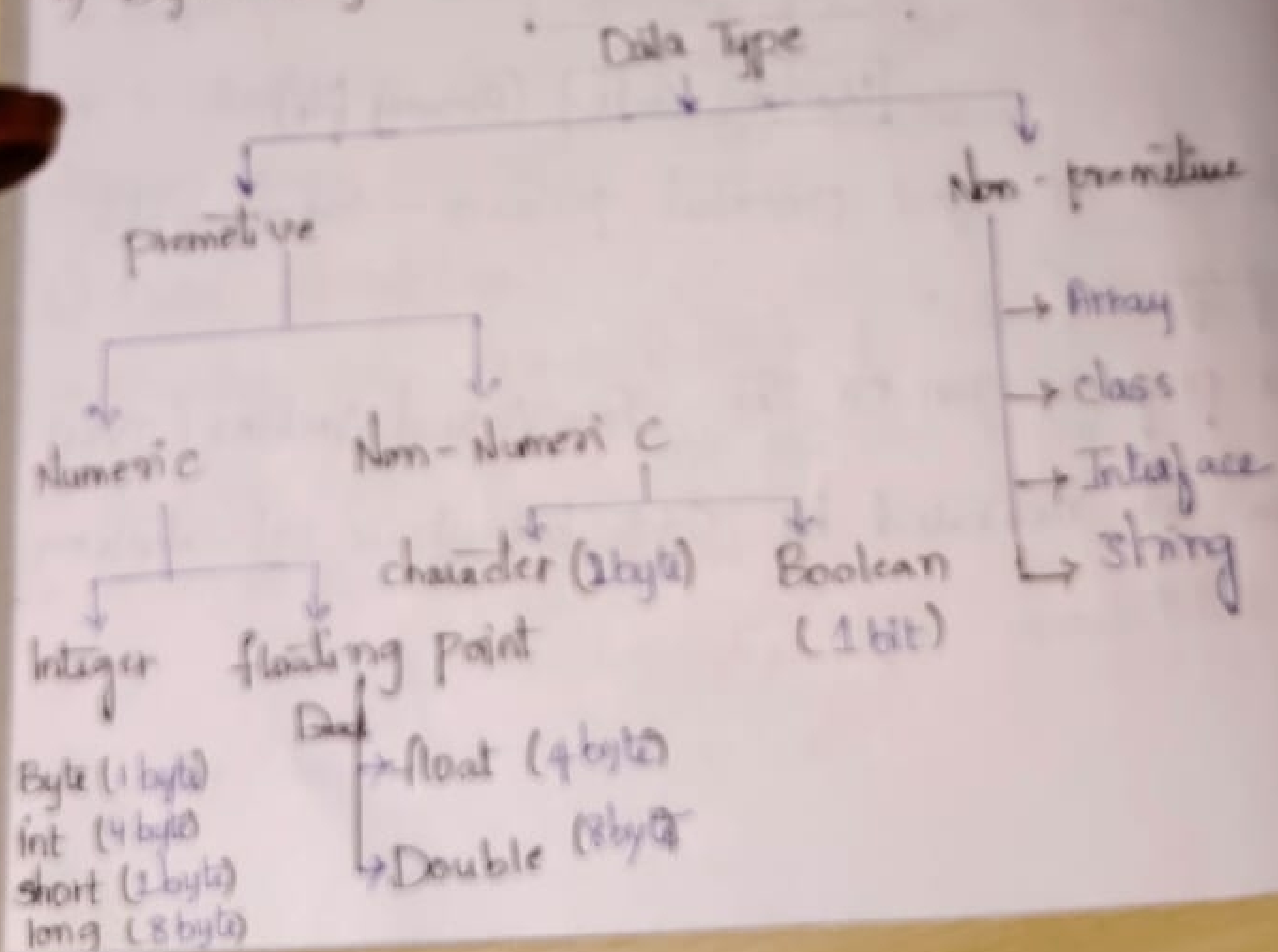
→ Java is an Object Oriented platform - Independent language.

→ Java programs run on the Java Virtual Machine (JVM).

→ Java got motivated by C++ but it is not extension of C++.

Java Requirements (or) Features of Java

- 1) Object Oriented
- 2) Compiled and Interpreted
- 3) Platform Independent & portable
- 4) Distributed
- 5) Robust & Secure
- 6) Small, Simple & popular. (It is small and simple because of code simplicity. As it becomes small & simple it becomes popular)
- 7) Scalability and performance.
- 8) Multi-Threaded. (Multi-tasks & needs to performed by one by one)
- 9) Interactive
- 10) High performance.
- 11) Dynamic & Extensible.



3) Inheritance

→ Inheritance promotes code reusability and reduces duplication.

→ We can use the same line of code multiple times.

Due to Inheritance of Java.

~~For~~ Inheritance is been adapted from the keyword 'extends'.

4) Encapsulation

→ Encapsulation is the process of combining data (Variables) and methods (functions) into a single unit.

Called a class

→ It protects the data from unauthorized access & improves security.

→ Encapsulation is a safeguard which keeps the data safe.

5) Polymorphism

→ Polymorphism means "many forms".

→ It allows the same method 'or' object to perform different tasks depending on the situation.

6) Data Abstraction

→ Showing only necessary details & hiding implementation.

→ It says what to do but not how to do.

→ It analyzes the problem.

→ It splits the big problem into small parts & solve it then combine it.

History

→ Developed

microsys

→ At that

'Java'

→ It is

→ Devel

1995

★
Hes